AI Playbook and Toolkit for Public Health Departments

AI Portfolio Strategy and Use Case Prioritization

EXE 410

AI opportunities can quickly exceed an agency's capacity to evaluate, implement, and sustain them. Leaders need a portfolio strategy that helps distinguish high-value public health uses from attractive but low-impact experiments. A portfolio approach organizes AI opportunities across programs, risk levels, readiness, equity impact, technical feasibility, expected benefit, and sustainability needs. This module teaches leaders how to create an AI use case intake and prioritization process. The module emphasizes that prioritization should not be based only on novelty or vendor availability. It should be based on mission alignment, public health value, data readiness, workflow fit, risk, equity, staff capacity, cost, and the ability to monitor and sustain the use case over time.

Level	advanced/leadership
Primary track	Public Health Executive Leadership Track
Appears in tracks	governance-security

Learning Objectives

- Define the purpose of an AI portfolio strategy for public health agencies.
- Develop criteria for prioritizing AI use cases based on public health value, readiness, risk, and sustainability.
- Identify use cases that should be pursued, piloted, deferred, redesigned, or rejected.
- Create a portfolio view that balances innovation, risk, equity, and operational capacity.
- Produce a use case prioritization rubric for governance or leadership review.

Module Overview

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This module teaches leaders how to create an AI use case intake and prioritization process. The module emphasizes that prioritization should not be based only on novelty or vendor availability. It should be based on mission alignment, public health value, data readiness, workflow fit, risk, equity, staff capacity, cost, and the ability to monitor and sustain the use case over time.

Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

Public health agencies have many possible AI use cases: summarizing case notes, analyzing call logs, drafting messages, identifying data quality gaps, forecasting demand, prioritizing inspections, supporting grants, and improving knowledge management. Without a portfolio strategy, agencies may choose projects because they are visible, easy to demonstrate, or championed by a strong vendor rather than because they advance the public health mission.

A portfolio strategy protects scarce capacity. Technical teams, privacy officers, legal counsel, procurement staff, program leaders, and data stewards cannot review every idea at the same intensity. Prioritization helps the agency focus deep review on use cases with meaningful benefit and material risk while creating faster pathways for low-risk internal productivity uses.

Public Health Example

A health department receives proposals for five AI uses: a public chatbot, a staff policy assistant, an outbreak signal detection model, a vendor tool to summarize eCRs, and an internal tool to draft grant narratives. Each proposal has different benefits and risks. A public chatbot may be visible but high-risk if it gives health advice. A staff policy assistant may be lower risk if it uses approved internal documents. An outbreak model may be valuable but require strong data quality and validation.

A portfolio review allows the agency to compare these proposals systematically. The agency may approve a low-risk internal pilot, defer the chatbot until public communication safeguards are developed, require validation planning for the outbreak model, and move the eCR summarization tool into deeper technical review. The result is a balanced portfolio rather than a series of disconnected decisions.

Technical and Governance Deep Dive

A useful portfolio strategy begins with categories. Agencies can classify AI opportunities by function, such as internal productivity, knowledge management, surveillance support, decision support, communication, automation, research, or public-facing service. They can also classify by risk, such as low, moderate, high, or prohibited. These categories help leadership understand where AI activity is concentrated and whether the portfolio reflects agency priorities.

Prioritization criteria should include mission alignment, expected public health benefit, workflow burden reduction, data readiness, legal authority, privacy sensitivity, equity impact, technical feasibility, implementation complexity, vendor dependency, cost, sustainability, and monitoring feasibility. A use case with high potential benefit but weak data readiness may be deferred until modernization work improves. A use case with moderate benefit and low risk may be an appropriate early pilot.

Portfolio governance should include regular review. Some use cases should move from idea to pilot, from pilot to operational use, or from operational use to retirement. Others should be stopped because they lack value, create unacceptable risk, duplicate another effort, or depend on unsustainable funding. A living portfolio helps leadership make these decisions transparently.

Risks, Failure Modes, and Guardrails

A major risk is pilot proliferation. Agencies may launch many small AI pilots without a pathway to scale, evaluate, sustain, or retire them. A guardrail is to require every pilot to define success criteria, evaluation evidence, funding assumptions, and an end point.

Another risk is inequitable prioritization. Projects serving better-resourced programs may advance faster than projects addressing underserved communities. A guardrail is to include equity impact and community benefit in the prioritization rubric.

Application to AI-Supported Workflows

Generative AI projects may be prioritized for internal knowledge management before public-facing use. Predictive analytics projects may be prioritized only when data and validation plans are mature. Agentic workflows may be deferred until governance has defined decision rights, access controls, and rollback procedures. Portfolio strategy helps leaders sequence AI adoption responsibly.

Reflection Questions

Which current or proposed AI use case in your agency raises this leadership or governance issue most clearly?

Who currently has authority to make decisions about this issue, and is that authority documented?

What evidence would governance or leadership need before approving the use case?

Which staff, partners, or communities would be affected if this issue were handled poorly?

What policy, process, or artifact should be created or updated after this module?

Practical Exercise

Develop an AI use case prioritization rubric with criteria, scoring levels, required evidence, and decision categories such as pursue, pilot, redesign, defer, or reject.

Before You Begin

Choose a real or realistic public health AI use case. Document assumptions, unresolved questions, and which agency roles would need to review the artifact before it is adopted. The exercise should produce a work product that could support governance review, leadership discussion, implementation planning, or policy development.

Expected Artifact or Evidence

Use case intake form; prioritization rubric; sample portfolio view; decision category definitions.

Expected Assignment

- Use case intake form; prioritization rubric; sample portfolio view; decision category definitions.

Knowledge Check

- 1. What is the strongest reason this module topic belongs in AI leadership and governance training?
- 2. Which practice best supports accountable public health AI governance?
- 3. When should an AI use case be escalated for leadership or governance review?
- 4. What is weak evidence of completion for a leadership and governance module?
- 5. What should leaders do when an AI use case has meaningful benefits but unresolved risks?

References and Resources

- NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0): https://www.nist.gov/itl/ai-risk-management-framework
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- NIST Privacy Framework
- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- WHO Ethics and Governance of Artificial Intelligence for Health: https://www.who.int/publications/i/item/9789240029200
- OMB Memorandum M-24-10 on federal AI governance, risk management, and innovation, where relevant to public-sector governance models
- Agency strategic plans, procurement policies, privacy policies, cybersecurity policies, records policies, and equity plans